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Multi-objective optimization for optimal placement of shared battery energy storage systems in urban energy communities

Jongbaek An [a], Taehoon Hong [b,*]

[a] Graduate Research Assistant and Ph.D. Candidate, Department of Architecture and Architectural Engineering, Yonsei University, Seoul 03722, South Korea
[b] Underwood Distinguished Professor, Department of Architecture and Architectural Engineering, Yonsei University, 50 Yonsei-ro, Seodaemun-gu, Seoul 03722, South Korea

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ABSTRACT

This study presents a novel multi-objective optimization approach for the optimal placement of shared battery energy storage systems (SBESS) in urban energy communities, balancing economic, technical, and environmental performance. Using real-world data from 191 buildings in Seoul, South Korea, a comprehensive model was developed that incorporates genetic algorithms and pareto optimality to optimize net present value, energy self-sufficiency rates, and peak loads. The results reveal that SBESS placement significantly improves technical and environmental performance compared to scenarios without SBESS, increasing energy self-sufficiency rates by up to 17.44% and reducing peak loads by up to 37.19% across different clusters. However, it currently lacks economic viability, with negative net present values ranging from USD -163,249 to USD -821,469 in all clusters. The optimal placement strategy achieved similar technical and environmental performances as the full placement approach yet required fewer installations. Sensitivity analysis emphasized the importance of electricity market design and policy frameworks in improving the economic feasibility of SBESS. This research contributes to urban energy planning by providing a realistic assessment of SBESS integration, offering valuable insights for policymakers and planners in balancing sustainability goals with economic constraints in energy community design.

1. Introduction

The seriousness of global warming caused by greenhouse gas emissions from human activities is becoming ever more noticeable. As a result, the international community has made various efforts to respond to the climate crisis. The transition from fossil fuel-based centralized energy production to energy production from green energy sources (e.g., new and renewable energy) has attracted global attention (Electricity 2024-Analysis and forecast to 2026, 2024). The International Energy Agency (IEA) estimated that photovoltaics (PV) accounted for 5% of global electricity generation in 2023, but the share would increase to 16% in 2030 due to the expansion of photovoltaics (PV) (Renewables 2024 - Analysis and forecast to 2030, 2024). However, electricity generation from renewable energy sources such as photovoltaic (PV) solar energy is greatly affected by climate and environmental conditions (An and Hong, 2024). In addition, intermittent and irregular renewable electricity generation can lead to unstable electricity supply and instability in an electricity system (Mahmud and Zahedi, 2016). Furthermore, a mismatch between electricity production and demand may make it difficult to efficiently utilize renewable energy. Some reports (California, 2012, Denholm et al., 2015) suggest that in California, the electricity load is very low during the day due to the overproduction of electricity from PVs, but the electricity load is still high at night when the PV system cannot produce electricity. The resulting huge electricity demand ramp is called 'the duck chart', named after its resemblance to a duck's neck. This can impose a heavy load on the electricity system. To solve this problem, the PV device connected to the electricity system must be removed. However, it is an inadequate solution at a point in time when the continuous expansion of renewable energy is required. In addition to the intermittency of PV generation, the electricity consumption of a building varies greatly depending on the building type, building occupancy, and building size.

Recently, battery energy storage systems (BESS) have been suggested as an effective solution to address the problems of renewable energy intermittency and electricity demand-supply mismatch. BESS has been proven to alleviate and solve various challenges presented (Hossain et al., 2023). First, BESS stores electricity produced by PV and discharges it when needed. This increases the stability of the electricity supply and enables effective load management (Rezaeimozafar et al., 2022, Günter

* Corresponding author.
E-mail address: hong7@yonsei.ac.kr (T. Hong).

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Nomenclature

BESS	battery energy storage system
ESSR	electricity self-sufficiency rate
GA	genetic algorithm
IEA	international energy agency
IIC	initial investment cost
LCC	life cycle cost
MoLIT	ministry of land, infrastructure and transport
NPV	net present value
PV	photovoltaic
REC	renewable energy certificate
SBESS	shared battery energy storage system
SMP	system marginal price
SOC	state of charge

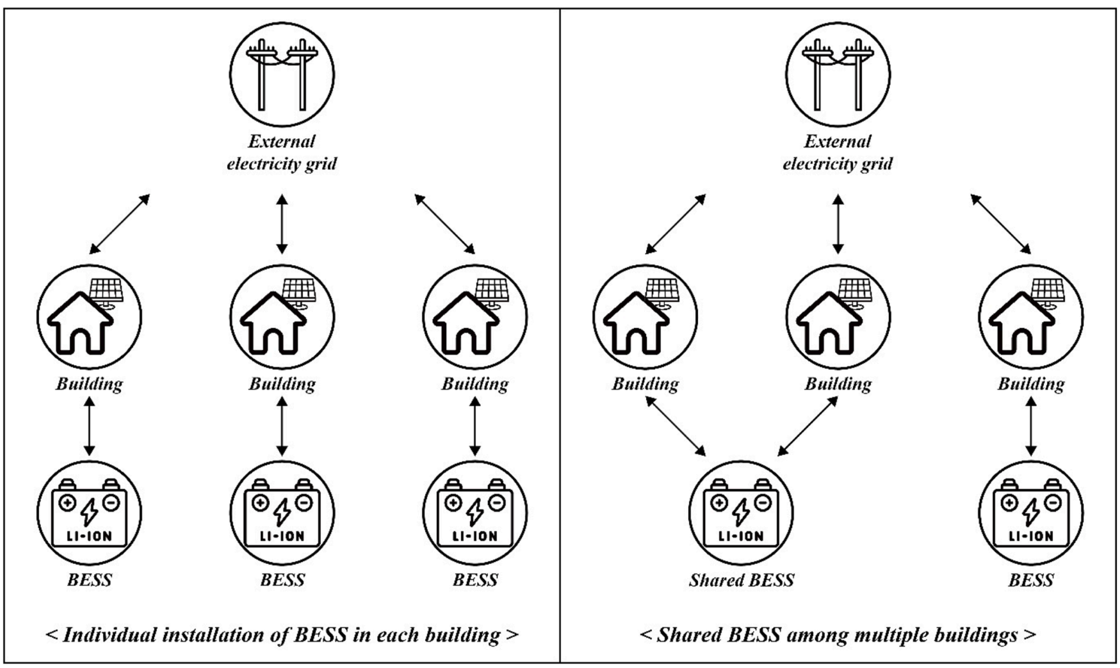


Fig. 1. Conceptual diagram of BESS installation methods.

and Marinopoulos, 2016, Yang et al., 2018). Second, BESS uses electricity produced by PV during the day to meet demand in the evening or at night, which can improve the self-consumption rate (Tan et al., 2021, Li et al., 2023). Finally, the strategic utilization of electricity through BESS can offer economic benefits. For example, BESS can be charged when electricity prices are low and then discharged when electricity prices are high due to a high load factor, thereby reducing the total cost of electricity (Masrur et al., 2024, Koohi-Fayegh and Rosen, 2020). In addition, BESS has various other advantages such as independence, rapid response to changes in demand, long-term energy management, increased reliability of electricity supply, and uninterrupted electricity supply (Cho et al., 2015, Nair and Garimella, 2010, Poullikkas, 2013, Luo et al., 2015). Therefore, BESS, which provides various technical advantages, is expected to further improve economic efficiency compared to a renewable energy system composed of only PV. However, the high installation cost of BESS still acts as a major barrier to its expansion.

There are two methods of installing BESS. The standard method is to install BESS in individual buildings. However, recently, due to the emergence of the share economy and the development of energy communities, a method of sharing BESS among multiple buildings has been examined in many studies (refer to Fig. 1).

One of the most important considerations in BESS installation in individual buildings is to determine the capacity of BESS. Many previous studies have been conducted to determine the optimal capacity of a standalone PV system with PV and BESS combined. Ahmad (2002), Bhuiyan and Asgar (2003), and Chel et al. (2009) performed a total life cycle cost analysis based on the daily electricity load and daily solar radiation data so as to calculate the optimal capacity of a standalone PV system. In these studies, the optimal capacity was derived based on economic viability, but other factors were not taken into account. Accordingly, it is difficult to say that optimization was performed in terms of technical and environmental aspects. Al-Salaymeh et al. (2010) calculated the optimal capacity of a standalone PV system based on data similar to the findings of previous studies, but failed to consider the technical and economic aspects, which may result in excessive capacity. As can be seen from the above cases, considering only a single objective or using an intuitive and simple method has the advantage of being able to quickly derive optimal capacity, but incorrect optimal capacity settings offset the benefits of BESS installation. Moreover, the installation of BESS with excessive capacity increases the initial investment costs (IIC), which then results in huge maintenance costs (Walker and Kwon, 2021).

The concept of shared battery energy system (SBESS) has emerged due to the high IIC for BESS installation in individual buildings (Liu et al., 2021, Roberts et al., 2019, Lüth et al., 2018). SBESS can reduce the economic burden by allowing multiple buildings to share BESS, thus sharing the expenses required for IIC, maintenance, and operation (Walker and Kwon, 2021, Dai et al., 2021). In addition, sharing BESS facilitates the exchange of electricity between buildings, which helps to reduce energy bills (Li et al., 2022). However, just like the BESS installation in individual buildings, capacity optimization is required for SBESS to ensure economic viability. Research seeking to determine the optimal capacity of SBESS has been conducted using various optimization technologies. Ma et al. (2022) proposed a method to simultaneously determine the optimal capacity and operation strategies of SBESS in electricity networks using a bi-level nested genetic algorithm. Hafiz et al. (2019) suggested a framework to manage energy in an energy community and determine the optimal capacity of SBESS through a multi-stage stochastic program model. This approach allowed for uncertainty so that more realistic SBESS operation strategies could be set up. Barbour et al. (2018) adopted a data-driven approach to analyze the economic feasibility of electricity storage using SBESS. The analysis found that the optimal capacity of SBESS is different from the sum of the capacities of BESS installed in individual buildings, which suggests that it is important to consider the electricity consumption of various buildings in determining the optimal capacity of SBESS. In the meantime, it is important to not only determine the optimal capacity but also to determine the optical placement of SBESS (Klaimi et al., 2018). Problems associated with the optimal placement of SBESS are dealt with based on the heuristic method (Mirtaheri et al., 2019, Prakash et al., 2022, Matthiss, Momenifarahani and Binder, 2021, Razavi et al., 2019, Wong and Ramachandaramurthy, 2020). Razavi et al. (2019) stated that the random placement of SBESS causes a huge loss in terms of electricity, and that optimal placement can improve efficiency in electricity. Ivanov et al. (2021) used genetic algorithms (GAs) to determine the optimal placement of SBESS so as to minimize electricity loss. Yunusov et al. (Zaidi, Bhatti and Ullah, 2018) analyzed the effects of the location and configuration of SBESS on peak load minimization, voltage and electricity loss in low-voltage electricity networks and proposed a method to derive optimal placement. Wong and Ramachandaramurthy (2020) used the whale optimization algorithm to determine the optimal placement of SBESS to minimize electricity loss. These studies commonly used metaheuristic techniques to perform optimization. However, most of them focused only on technical and environmental performance (i.e., self-sufficiency rate, peak load, electricity losses, etc…) and did not consider economic performance.

Therefore, while existing studies have used various objectives and methodologies to optimize BESS installed individual buildings and energy communities, there is still a lack of research related to the optimal placement for energy communities. In addition, previous studies considered only a single objective or purpose with similar aspects, and research related to the optimal placement considering various objectives has not yet been performed. Therefore, this study aims to derive the optimal placement considering various optimization objectives when installing and sharing BESS within the energy community: (i) economic performance (i.e., net present value (NPV)); and (ii) technical and environmental performance (i.e., electricity self-sufficiency rate (ESSR), and peak load). Based on previous research, independent variables for deriving the optimal placement were set as follows: (i) capacity of SBESS; and (ii) location of SBESS.

This study makes the following main contributions: (i) It adopts a multi-objective optimization approach to balance economic, technical, and environmental performance, providing a comprehensive evaluation framework for SBESS placement; (ii) It utilizes actual energy community data, increasing the practicality and reliability of the results compared to previous studies that often relied on hypothetical scenarios; (iii) It incorporates practical factors such as transmission losses, capacity deterioration, self-discharge, and charge/discharge efficiency, thereby improving the realism and applicability of the proposed model.

2. Materials and methods

2.1. Stage 1: selection of the study area

This study sought to analyze the effects of SBESS installed together with PV in energy communities in urban areas in terms of economic, technical, and environmental performance. To this end, the study area was selected based on the following criteria.

- Energy consumption: Areas with high energy consumption have great potential to increase ESSR due to the installation of renewable energy systems and are important research targets for seeking efficient energy management methods.
- Renewable energy deployment rate: Renewable energy deployment rates are an important indicator for identifying the potential and opportunities through the introduction of renewable energy systems in the target area. When the renewable energy deployment rate is low, the area has a high possibility of increasing ESSR and energy sustainability through the spread of renewable energy systems. Even when the renewable energy deployment rate is high, there is room for improving energy efficiency through further improvement in efficiency and system optimization. Thus, by considering the renewable energy deployment rate, it is possible to assess the region's potential for energy transition. This insight presents a critical research opportunity to explore strategic approaches for the implementation of sustainable energy systems.
- Mixture of various building types: Generally, in urban areas, various types of buildings such as residential, commercial, public, and educational institutions are mixed together. Because rooftop areas and energy consumption patterns vary depending on the type of building, consideration of electricity production and consumption patterns for each type of building is a key factor in promoting the efficient use of renewable energy throughout the city.
- Population density: Areas with high population density tend to have relatively higher energy consumption, making it crucial to secure energy supply stability and improve energy efficiency. High population density areas provide a suitable environment for evaluating and applying such solutions effectively.

In consideration of the above factors, Daechi 1 dong, one of the administrative districts of Seoul, South Korea, was selected as the study area. Daechi 1 dong is located in Gangnam-gu and has the highest energy consumption in Seoul. It also has complex urban characteristics, with various building types (i.e., apartments, commercial and business buildings, public buildings, and schools) (Korea real estate board, 2024). In addition, it has the highest population density (i.e., 30,220 people/km^2) in Gangnam-gu (KOSIS KOrean Statistical Information Service, 2024), providing an environment suitable for urban energy pattern analysis and research on renewable energy utilization methods. The relatively low renewable energy deployment rate compared to the high energy consumption of Daechi 1 dong suggests high potential in terms of the diffusion of renewable energy (Korea New and Renewable Energy Center, 2024). In this study, it is assumed that PV, which has the greatest potential among renewable energy sources and is easy to integrate with buildings, is installed in urban areas, when the economic, technical, and environmental performance according to the placement of SBESS is to be analyzed. In this vein, the characteristics of Daechi 1 dong provide ideal conditions to perform multi-objective optimization research for the optimal placement of PV and SBESS. The target area has various building types and energy consumption patterns, making it an appropriate study area for analyzing economic, technical, and environmental performance in a comprehensive manner.

2.2. Stage 2: data collection

2.2.1. Database on energy and physical characteristics of buildings

A total of 191 buildings are located in the study area (i.e., Daechi 1 dong). In this study, a database on the energy consumption and physical characteristics of these buildings were built and used for analysis (refer to Table 1). First, the energy consumption was collected from the Korea Real Estate Board, which collects and manages information related to the characteristics of real estate (Korea real estate board, 2024), and data related to the energy consumption of education facilities was collected from reports published by the Ministry of Education (2024). Second, the physical characteristics, including coordinates, rooftop areas, building heights and altitude above sea level, were collected from GIS building integrated information provided by V-WORLD under the

Ministry of Land, Infrastructure and Transport (MoLIT) and digital topographic maps provided by the National Geographic Information Institute (V-world open market, National Geographic Information Institute, 2024).

The collected data is used not only to estimate the electricity generation of PV installed on the rooftop of a building, but also to calculate transmission loss in electricity transmission due to the installation location and physical distance of SBESS.

2.2.2. Database on PV ans SBESS

Data on PV modules and solar radiation were collected to calculate the electricity generation of PV installed on building rooftops in the study area. First, in relation to information about PV modules, the Q. PEAK-G5 300 watt produced by Hanwha Qcells, which has the highest market share in South Korea, was selected via market research (KOSIS KOrean Statistical Information Service, 2024, US distributed solar and storage competitive landscapes shift in 2023, 2024) (refer to Table 2). Second, solar radiation was collected by the World Radiation Data Centre sponsored by the World Meteorological Organization, when the average hourly solar radiation for five years from 2019 to 2023 were collected (World Radiation Data Centre, 2024). Third, the azimuth and altitude of the Sun change every hour. Therefore, in urban areas with densely packed buildings with different heights, the Sun's azimuth and altitude must be collected every hour to accurately estimate the electricity generation of PV with the effects of shadows reflected. Korea Astronomy & Space Science Institute, which conducts research and development related to astronomy, presents a calculator that calculates the azimuth and altitude of the Sun according to date and time (Korea Astronomy and Space Science Institute, 2024). In this study, the azimuth and altitude of the Sun from sunrise to sunset on the 15th day of every month were collected to reflect the average environment. Based on the collected data, the actual electricity generation of rooftop PV with the characteristics of cities reflected can be calculated using the hillshade tool of ArcGIS. In the previous research, the authors of this study estimated the electricity generation of PV with the shadow effect in the rooftop area of a building considered, and the detailed prediction process was not described to avoid repeating the same content. For a detailed description of a method to estimate the electricity generation of rooftop PV considering the surrounding environment, please refer to An and Hong (2024).

The installation of SBESS is essential to efficiently utilize the electricity produced by PV. Unlike PV, which is widely distributed due to its stabilization in terms of price and performance, SBESS is subject to uncertainty in terms of technology and economic viability. And due to its high IIC, SBESS is installed only in limited cases in South Korea. In addition, because it is difficult to investigate the market price, the price according to the capacity of SBESS was set by referring to the report published by IEA (IEA), and information related to the performance of SBESS was set based on existing research and manufacturers' catalogs (LG home battery product info, 2024, Aneke and Wang, 2016) (refer to Table 2).

2.2.3. Database on economic feasibility

This study sought to evaluate the economic performance of SBESS combined with PV over time based on a life cycle cost (LCC) analysis. To this end, the following data were collected, and resulting assumptions were made: (i) analysis approach; (ii) analysis period; (iii) real discount rate; (iv) electricity trading prices; and (v) significant costs of ownership. First, the present worth method was determined as an analysis approach to reflect changes in cash value with time. Second, considering that reinforced concrete is the predominant structural material in modern buildings, the analysis period was set to 40 years (Korean Law Information Center, 2024). This duration corresponds to the legal durability period for buildings in South Korea, reflecting the expected lifespan of such structures. Third, real discount rates were collected to reflect changes in cash value. The real discount rate calculation requires nominal interest rates and inflation rates, which were collected from the Economic Statistics System operated by the Bank of Korea (Economic Statistics System, 2024). In this study, data for the past decade were collected to prevent rapid changes in a short period of time, and the average nominal interest rate was 2.07%, and the inflation rate 1.66%. Fourth, the system marginal price (SMP), renewable energy certificate (REC) price (Korea power exchange, 2024) and electricity price (Korea Electric Power COrporation, 2024) were collected to calculate the revenue from electricity produced by PV. While SMP was collected based on the time of day (i.e., USD 0.1/kWh ~ USD 0.13/kWh), electricity price was collected according to season (i.e., spring and fall: 0.07/kWh, summer: USD 0.1/kWh, and winter: USD 0.09/kWh). The REC price was USD 0.06/kWh. When electricity produced by PV is sold to an external electricity grid, the price is set in the form of SMP+REC. Finally, the significant cost of ownership includes the IIC, maintenance costs, and replacement costs of PV and SBESS as cost items, and electricity bill saving as a profit item, when electricity bill saving refers to the revenue from the sale or self-consumption of electricity produced by PV.

2.3. Stage 3: PV-SBESS system modeling

2.3.1. Basic electric system modeling

In order to evaluate the performance of the PV-SBESS system, there is a need for a modeling of an electric system capable of simulating the flow of electricity. In this study, a basic electric system consisting of a building, PV, external electricity grids, and SBESS was defined based on the following equation (refer to Fig. 2) (Kang et al., 2023, Jung et al., 2020).

The electricity produced by PV and the electricity consumed by the building are defined as in Eqs. (1) and (2). The charge and discharge quantities of ESS are defined as in Eqs. (3) and (4). The amounts of electricity purchased or sold from the external electricity grid are defined as in Eqs. (5) and (6). Finally, the supply of electricity has to match the electricity load to achieve power balance in the electric system (refer to Eq. (7)). Additionally, for SBESS, transmission losses due to the physical distance between the PV that produces electricity and the SBESS that stores it were considered as in Eq. (8).

$$E_t^{PV} = E_t^{P2B} + E_t^{P2E} + E_t^{P2G} \tag{1}$$

$$E_t^{B} = E_t^{P2B} + E_t^{E2B} + E_t^{G2B} \tag{2}$$

$$E_{charge,t}^{SBESS} = E_t^{P2E} \tag{3}$$

$$E_{discharge,t}^{SBESS} = E_t^{E2B} + E_t^{E2G} \tag{4}$$

$$E_{import,t}^{Grid} = E_t^{G2B} \tag{5}$$

$$E_{export,t}^{Grid} = E_t^{P2G} + E_t^{E2G} \tag{6}$$

$$E_t^{PV} + E_{discharge,t}^{SBESS} + E_{import,t}^{Grid} - \left(E_t^{B} + E_{charge,t}^{SBESS} + E_{export,t}^{Grid}\right) = 0 \tag{7}$$

$$E_{loss} = \left(\frac{P}{V}\right)^2 \times \rho \times \frac{D}{A} \tag{8}$$

where, $\mathrm{E_t^{PV}}$ stand for the amount of electricity generated by PV at time t, $\mathrm{E_t^{P2B}}$ stand for the amount of PV generated electricity self-consumed at time t, $\mathrm{E_t^{P2E}}$ stand for the amount of PV generated electricity charged to SBESS at time t, $\mathrm{E_t^{P2G}}$ stand for the amount of PV generated electricity sold to the external electricity grid at time t, $\mathrm{E_t^{B}}$ stand for the electricity load of the building at time t, $\mathrm{E_t^{P2B}}$ stand for amount of PV generated electricity self-consumed at time t, $\mathrm{E_t^{E2B}}$ stand for the amount of electricity discharged from SBESS to the building at time t, $\mathrm{E_t^{G2B}}$ stand for the amount of electricity consumed from purchases from the external electricity grid at time t, $\mathrm{E_{charge,t}^{SBESS}}$ stand for the amount of electricity charged

Table 1
Energy and physical characteristics of buildings within the study area.

Classification	Detailed classification	Unit	Refs.
Energy	Electricity consumption	kWh	(Korea real estate board, 2024, Ministry of Education, 2024)
Physical	Coordinate	-	(V-world open market, 2024)
	Rooftop area	m^2	(V-world open market, 2024)
	Building height	m	(V-world open market, 2024)
	Altitude above sea level	m	(National Geographic Information Institute, 2024)

to SBESS at time t, $E^{SBESS}_{discharge,t}$ stand for the amount of electricity discharged from SBESS at time t, $E^{Grid}_{import,t}$ stand for the amount of electricity imported from the external electricity grid at time t, $E^{Grid}_{export,t}$ stand for the amount of electricity discharged to the external electricity grid at time t, E_{loss} stand for transmission losses based on distance, P stand for transmitted electricity, v stand for voltage, ρ stand for resistance of the transmission line, D stand for distance, and A stand for cross-sectional area of the transmission line.

2.3.2. Shared battery energy storage system modeling

It is not only necessary to implement the electricity flow of the entire electric system, it is also important for SBESS to operate properly. The state of charge (SOC), which means the amount of electricity stored in SBESS, can be calculated using Eq. (9) (Kang et al., 2023, Jung et al., 2020). In this study, the SOC of SBESS is updated considering the charge/discharge efficiency. Furthermore, in order to prevent the performance degradation of SBESS due to excessive charge/discharge quantity, the minimum and maximum SOC were set to limit the charge/discharge quantity (refer to Eq. (10)). Eq. (11) requires that SBESS should not be charged and discharged simultaneously, and the charge/discharge capacity of SBESS is calculated based on the amount of electricity stored in SBESS (refer to Eqs. (12) and (13)). Finally, the capacity of SBESS, which deteriorates every year, is calculated based on the preset degradation rate (refer to Eq. (14)).

$$SOC_t = SOC_{t-1} \times \left(1 - \eta_{self.d}\right)^{\Delta t} + \frac{E^{SBESS}_{charge,t} \times \eta_c - E^{SBESS}_{discharge,t} \times \eta_d}{Capa_{SBESS}} \tag{9}$$

$$SOC_{min} \le SOC_t \le SOC_{max} \tag{10}$$

$$E^{SBESS}_{charge,t} \times E^{SBESS}_{discharge,t} = 0 \tag{11}$$

$$0 \le E^{SBESS}_{charge,t} \le P_{charge,max} \tag{12}$$

$$0 \le E^{SBESS}_{discharge,t} \le P_{discharge,max} \tag{13}$$

$$Capa_{degraded,t} = Capa_{initial} \times \left(1 - r_{degradation}\right)^t \tag{14}$$

where, SOC_t stand for state of charge at time t, $\eta_{self.d}$ stand for self-discharge rate at time t, η_c stand for charge efficiency at time t, η_d stand for discharge efficiency at time t, $E^{SBESS}_{charge,t}$ stand for the amount of electricity charged to SBESS at time t, $E^{SBESS}_{discharge,t}$ stand for the amount of electricity discharged from SBESS at time t, $Capa_{SBESS}$ stand for capacity of SBESS, SOC_{min} stand for minimum state of charge, SOC_{max} stand for maximum state of charge, Pcharge,max stand for maximum charging electricity of SBESS, Pdischarge,max stand for maximum discharging electricity of SBESS, $Capa_{degraded,t}$ stand for degraded SBESS capacity at time t, $Capa_{initial}$ stand for initial SBESS capacity, $r_{degradation}$ stand for degradation rate of SBESS.

2.4. Stage 4: definition of the optimization goal for the optimal placement of SBESS

2.4.1. Metric for economic performance evaluation

The economic performance of SBESS combined with PV was analyzed using NPV, an absolute indicator of the present worth method (Jeong et al., 2022, An et al., 2019). Generally, NPV≥0 indicates that it is economically feasible. In this study, NPV was calculated in the following manner:

First, the cost items of PV and SBESS can be calculated using Eqs. (15)–(18).

$$IIC = IIC_{PV} + \sum_{i=1}^{n_{SBESS}} IIC_{SBESS,i} \tag{15}$$

$$Maintenance_t = \frac{IIC_{PV} \times MC_{PV} + \sum_{i=1}^{n_{SBESS}} IIC_{SBESS,i} \times MC_{SBESS}}{(1+r)^t} \tag{16}$$

$$Replacement_{PV,t} = \frac{IIC_{PV,i}}{(1+r)^t},\ t = k \times T_{PV,life}\ (k = 1,\ 2,\ 3, \ldots) \tag{17}$$

$$Replacement_{SBESS,t} = \sum_{i=1}^{n_{SBESS}} \frac{IIC_{SBESS,i}}{(1+r)^t},\ t = k \times T_{SBESS,life}\ (k = 1,\ 2,\ 3, \ldots) \tag{18}$$

where, IIC stands for initial investment cost, IIC_{PV} stands for initial investment cost of PV, IIC_{SBESS} stands for initial investment cost of SBESS, n_{SBESS} stands for number of SBESS, MC_{PV} stands for maintenance cost of PV, MC_{SBESS} stands for maintenance cost of SBESS, $T_{PV,life}$ stands for useful life of PV, $T_{SBESS,life}$ stands for useful life of SBESS, r stands for real discount rate, t stands for year.

Second, the revenue items of PV and SBESS can be calculated using Eqs. (19)–(21) (Korea power exchange, 2024, Korea Electric Power COrporation, 2024, Kim et al., 2023).

$$EBS_t = \sum_{h=1}^{24} \left(E^{E2B}_{h,t} + E^{P2B}_{h,t}\right) \times P_h \tag{19}$$

$$Revenue_t = \sum_{h=1}^{24} \left(E^{E2G}_{h,t} + E^{P2G}_{h,t}\right) \times (SMP_h + REC) \tag{20}$$

$$Total\ revenue_t = \frac{EBS_t + Revenue_t}{(1+r)^t} \tag{21}$$

where, EBS_t stands for electricity bill saving at year t, $E^{E2B}_{h,t}$ stands for electricity self-consumed from the SBESS at time h and year t, $E^{P2B}_{h,t}$ stands for electricity self-consumed from the PV at time h and year t, P_h stands for electricity price at time h, $Revenue_t$ stands for revenue from electricity sales at year t, $E^{E2G}_{h,t}$ stands for electricity sold from the SBESS to the grid at time h and year t, $E^{P2G}_{h,t}$ stands for electricity sold from the PV to the grid at time h and year t, SMP_h stand for SMP at time h, REC stands for REC price, Total $revenue_t$ stands for total revenue from self-consumption and sales of electricity at year t, r stands for real discount rate, t stands for year.

Finally, NPV can be calculated using Eq. (22).

$$NPV_t = -IIC - \sum_{t=1}^{T} \left(Maintenance_t + Replacement_{PV,t} + Replacement_{SBESS,t}\right) + \sum_{t=1}^{T} Total\ revenue_t \tag{22}$$

where, NPV_i stands for net present value at year t.

Table 2
Physical and economic characteristics of PV and SBESS.

Classification	Classification	Description
Photovoltaic	Module power	300W
	Module efficiency	97%
	Useful life	25years
	IIC_{PV}	USD 362.47/module
	$O\&M_{PV}$	1% of IIC_{PV}/year
	RC_{PV}	9.5% of IIC_{PV}/every 13year
	Miscellaneous losses	3%
	Annual degradation rate	0.8%
Shared battery energy storage system	System capacity	3.3, 9.8, 16, 20, 24 kWh
	State of Charge	20~90%
	Charge/discharge efficiency	95%
	Self-discharge rate	5%/month
	Useful life	10 years
	IIC_{BESS}	USD 235.29/kWh
	$O\&M_{BESS}$	2.2%/year
	Annual degradation rate	0.21%/year

Note: The exchange rate (KRW/USD) is 1383.11 won to a U.S. dollar (as of 23 July 2024), IIC_{PV} stand for initial investment cost of PV, $O\&M_{PV}$ stand for operation and maintenance cost of PV, RC_{PV} stand for replacement cost of PV inverter, IIC_{SBESS} stand for initial investment cost of SBESS, $O\&M_{SBESS}$ stand for operation and maintenance cost of SBESS.

2.4.2. Metrics for technical and environmental performance evaluation

The technical and environmental performance of SBESS combined with PV can be evaluated through ESSR and peak load. First, the environmental performance of the system with PV and SBESS combined was evaluated using ESSR. ESSR is a quantification of the self-consumption rate of electricity produced in a renewable energy system and indicates its utilization efficiency. In the renewable energy system, high ESSR means a high self-consumption rate of the electricity produced, which leads to a reduction in dependence on the external electricity grid. This reduced dependence in turn contributes to reducing carbon emissions, especially in areas with a fossil fuel-based electricity supply, and thereby maximizes the environmental benefits of the PV-SBESS system. In addition, regions with high ESSR can maintain a stable supply through electricity self-efficiency even when the external electricity grid is unstable, or electricity outages occur, which helps improve energy resilience. ESSR is calculated by dividing the total amount of PV and SBESS electricity self-consumed by the system over a specific period by the total electricity demand. ESSR can be calculated using Eq. (23).

$$ESSR = \frac{\sum_{t=1}^{T}\left(E_t^{P2B} + E_t^{E2B}\right)}{\sum_{t=1}^{T} EC_{total,t}} \tag{23}$$

where, ESSR stands for energy self-sufficiency rate, $E_{h,t}^{P2B}$ stands for electricity self-consumed from the PV at time t, $E_{h,t}^{E2B}$ stands for electricity self-consumed from the SBESS at time t, $EC_{total,t}$ stands for total electricity load of the system at time t, and t stand for years.

Second, peak load is a crucial metric for evaluating the technical performance of PV-SBESS systems, representing the maximum elec-

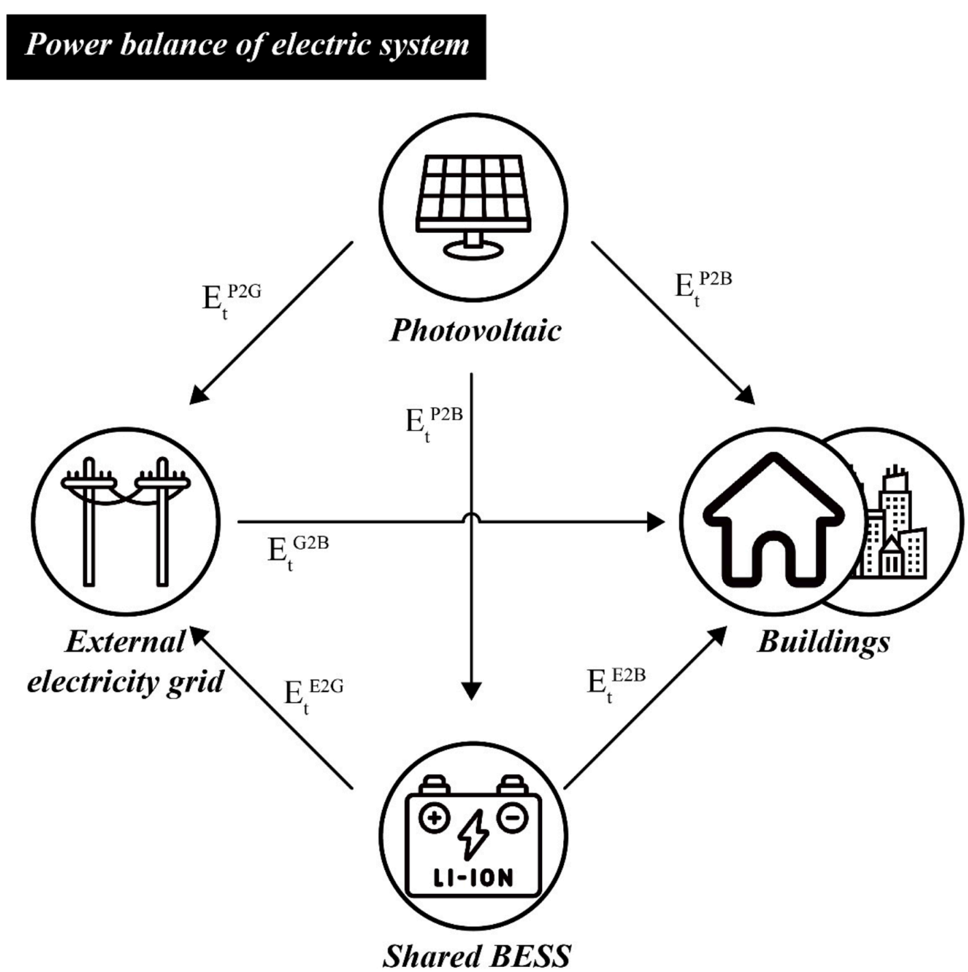


Fig. 2. Power balance within the electric system.

Table 3
Researches on BESS placement optimization using heuristic optimization methods.

Refs.	Author	Optimized variables		Optimization functions	Optimization method
		Location	Capacity		
(Mirtaheri et al., 2019)	Mirtaheri et al.	O	O	• Voltage deviation • Power losses	• Genetic algorithm
(Prakash et al., 2022)	Prakash et al.	O	O	• Overvoltage • Voltage fluctuations • BESS costs	• Particle Swarm Optimization algorithm
(Matthiss, Momenifarahani and Binder, 2021)	Matthiss et al.	O	O	• Energy Storage Costs • Grid Reinforcement Costs	• Heuristic grid reinforcement algorithm
(Wong and Ramachandaramurthy, 2020)	Wong and Ramachandaramurthy	O	O	• Energy losses	• Whale Optimization algorithm
(Saini and Gidwani, 2022)	Saini and Gidwani	O	O	• Energy losses • Overvoltage • Reverse power flow	• Genetic algorithm
(Huang, Zhang and Xiao, 2010)	Huang et al.	O		• Voltage stability index	• Genetic algorithm
(Vargas and Jiménez-Estévez, 2011)	Vargas and Jiménez-Estévez	O	O	• Energy losses • Voltage drop	• Genetic algorithm
(Leou, 2012)	Leou		O	• Economic performance	• Genetic algorithm
(An and Hong, 2025)	An and Hong	O	O	• Economic performance • Peak load • Electricity self-sufficiency rate	• Genetic algorithm

tricity consumption in a specific area during a given time period. Specifically, it is defined as the maximum value of electricity imported from the external electricity grid during peak hours, when electricity consumption is at its highest. Peak load analysis allows for quantitative assessment of how the PV-SBESS system manages electricity during maximum load times and its dependence on the external electricity grid. A lower peak load value indicates that the PV-SBESS system is self-supplying more electricity during peak hours, leading to improved electricity self-sufficiency and reduced reliance on the external electricity grid. Consequently, peak load serves as a key metric for assessing the efficiency and autonomy of PV-SBESS systems, playing a vital role in system design and operational optimization. Peak load can be calculated using Eq. (24).

$$Peak\ load = \max_{t \in \text{peak hours}} \left(\sum_{n=1}^{N} Import_{n,t} \right) \tag{24}$$

where, N stand for number of building, $import_{n,t}$ stand for the amount of electricity imported from the external electricity grid by building n at time t, peak hours stand for the time periods of highest electricity load.

2.5. Stage 5: optimal SBESS placement using multi-objective optimization

In order to determine the optimal SBESS placement, a multi-objective optimization was performed considering the optimization objectives (i.e., NPV, ESSR, and peak load) defined in stage 4. As can be seen in Section 2.2.3.the current electricity price (i.e., USD 0.07~0.1/kWh) in South Korea is significantly lower than the SMP and REC prices (i.e., USD 0.16~0.19/kWh). That is, as the electricity produced by PV is used more frequently for self-consumption, ESSR increases, and peak load decreases. This, in turn, increases technical and environmental performance, but the revenue that can be obtained decreases, resulting in a reduction of NPV in terms of economic performance. In other words, ESSR and peak load, and NPV are in a trade-off relationship. When there is a trade-off between two or more conflicting optimization goals, multiple pareto solutions are present. Therefore, it is important to determine the optimal SBESS placement in consideration of a balance between the optimization goals. For this purpose, GA and Pareto optimality were used in this study. GA is one of the most well-known iterative heuristic optimization methods based on biological evolution (Mirjalili, 2019, Grefenstette, 1986). Although GA is considered a traditional algorithm, it remains a powerful methodology that is widely used today to address problems related to determining the location, type, and capacity of BESS in electricity grids (refer to Table 3) (Mirtaheri et al., 2019, Prakash et al., 2022, Matthiss et al., 2021, Wong and Ramachandaramurthy, 2020, Saini and Gidwani, 2022, Huang et al., 2010, Vargas and Jiménez-Estévez, 2011, Leou, 2012, An and Hong, 2025). Heuristic approaches, such as GA, offer advantages including fast convergence, ease of application, and high adaptability (Yang et al., 2018, Houck et al., 1995). Moreover, heuristic optimization methods are particularly effective for addressing highly complex problems involving nonlinear or disjunctive constraints, as they can operate without relying on intricate mathematical formulations. However, heuristic optimization methods are not global optimization techniques, and thus they do not guarantee the derivation of a global optimal solution. Additionally, the time required to derive an optimal solution can vary significantly depending on the complexity of the problem being addressed and the configured hyperparameters (Grisales-Noreña et al., 2024).

The model, which is aimed to determine the optimal SBESS placement with the concept of GA and Pareto optimality applied, consists of three steps: (i) hyperplane; (ii) normalization; and (iii) fitness function (An et al., 2019, Konak et al., 2006, Alabsi and Naoum, 2012, Koo et al., 2015). First, the hyperplane is defined based on the minimum and maximum extreme points of each optimization goal. Second, all optimization goals are converted into normalized values between 0 and 1 so that they can be compared on the same scale. Third, the fitness function for performing multi-objective optimization is defined using the weighted Euclidean distance (refer to Eq. (25) and Table 4).

$$Fitness\ function = \sqrt{w_A \times (1 - N_A)^2 + w_B \times (1 - N_B)^2 + w_C \times (N_C - 0)^2} \tag{25}$$

where, w_A, w_B, and w_C stand for weight factor of optimization objectives A, B, and C, N_A, N_B, and N_C stand for normalized value of optimization objective A, B, and C.

NPV and ESSR with the optimization goal of maximization are shown in the form of $(1\text{-}N_A)^2$ because the closer they are to 1, the better the performance. On the other hand, peak load with the optimization goal of minimization is shown in the form of $(N_C\text{-}0)^2$ because the closer it is to 0, the better the performance. In addition, when the value of the fitness function reaches the minimum value, it is seen as the optimal solution, when the SBESS placement becomes the optimal SBESS placement. Meanwhile, in this study, GA was used for capacity design and optimal placement of SBESS. In GA, an individual, which contains information related to independent variables, is composed of binary (0, 1) and

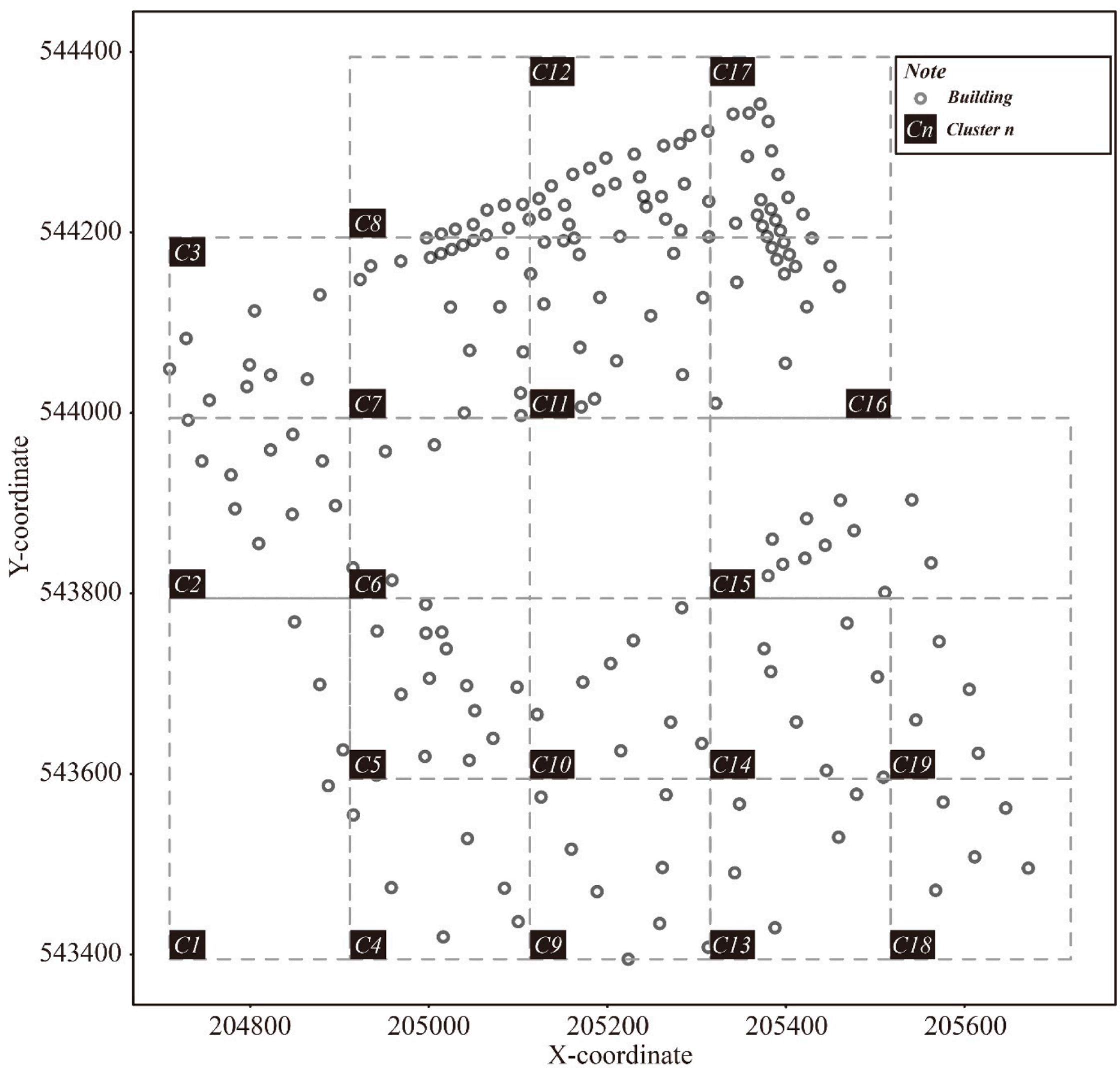
Table 4
Optimization objectives and goals for optimal SBESS placement.

Classification	Perspective	Optimization objectives	Optimization goals
Goal A	Economic	NPV	Maximization
Goal B	Technical and environmental	ESSR	Maximization
Goal C		Peak load	Minimization

integer (1, 2, 3, 4, 5) tuples that indicate the installation of SBESS in a building and the type. In the individual, each element contains a type index that determines (i) the installation of SBESS and (ii) the capacity of SBESS. For example, a tuple (1, 2) indicates that type 2 SBESS is installed in the node. In this manner, each individual represents the SBESS placement and capacity for multiple buildings in the electric system, and searches for the optimal solution through crossover and mutation, which are basic operators in GA. The crossover operator takes the parts of two individuals to create a new individual, while the mutation operator alters the SBESS installation status and SBESS type according to a certain probability. Through the iterative search of GA, the optimal combination of SBESS placement and capacity in the electric system was found, and the optimization was performed considering the goals of increasing ESSR and NPV and reducing peak load in a comprehensive manner.

3. Results and discussion

In this study, grid-based clustering was performed for the optimal placement of SBESS in a large urban area (refer to Fig. 3). In the study area, which is a wide administrative district with the width and breadth of 1km, this approach provides several significant advantages. First, it greatly improves computational efficiency, allowing for the relatively fast optimal placement of a large PV-SBESS system. Second, it limits the distance between SBESS and buildings in each cluster to minimize transmission losses, thereby improving the overall efficiency of the PV-SBESS system. Third, it reflects geographical proximity and can thus consider the physical constraints of the actual electricity network. In addition, this method can be consistently applied, even to large areas, providing high scalability and opening up the possibility of parallel processing through the independent optimization of each cluster.



Note: The coordinates system are expressed in meters based on the Korea Central Belt.
The X-axis represents the west-east direction, while the Y-axis represents the south-north direction.
On the coordinate plane, 1 unit represents 1 meter.

Fig. 3. Result of grid based clustering.

Table 5
Evaluation of optimization objectives according to optimal placement of SBESS by cluster.

Cluster ID	ESSR	Peak load	NPV	Number of SBESS				
				Type 1	Type 2	Type 3	Type 4	Type 5
-	(%)	(kWh)	(USD)	(N)	(N)	(N)	(N)	(N)
1	4.95	291.22	-322,134.11	1		1		
2	5	425.06	-501,199.90		1	4		
3	3.33	749.35	-487,120.95		1	1	1	
4	7.58	333.17	-531,644.24	1	1			1
5	4.36	583.42	-632,275.84	1	1	1	1	
6	4.27	305.91	-304,278.44				3	
7	8.2	439.96	-821,469.37		1	3	1	
8	3.28	215.99	-163,249.56			1		
9	10.52	211.16	-484,829.16	2	1			1
10	5.31	668.81	-741,780.22		2	1		
11	5.74	347.46	-538,878.16	1	3	2		
12	3.94	674.5	-638,256.10			2	3	1
13	8.74	212.82	-358,821.84		2			
14	4.72	659.83	-719,358.79	1	1			
15	2.97	1102.09	-725,029.68	1	2	1		
16	4.27	481.05	-584,262.89	1	2	2	2	
17	4.51	283.4	-353,131.25					
18	7.96	196.15	-330,977.89	1			2	1
19	17.44	92.39	-284,713.65	1				

Note: The exchange rate (KRW/USD) is 1383.11 won to a U.S. dollar (as of 23 July 2024), Type 1 stand for SBESS with a capacity of 24 kWh, Type 2 stand for SBESS with a capacity of 20 kWh, Type 3 stand for SBESS with a capacity of 16 kWh, Type 4 stand for SBESS with a capacity of 9.8 kWh, Type 5 stand for SBESS with a capacity of 3.3 kWh.

Finally, geographically separated clusters can improve the efficiency of maintenance and management when implementing actual PV-SBESS systems. Due to these various advantages, grid-based clustering can be considered an effective and practical approach for optimizing large-scale urban energy systems. Based on the grid-based clustering results, multi-objective optimization was performed considering NPV, ESSR, and peak load according to the placement of SBESS. GA generates various SBESS placements through crossover and mutation, calculates NPV, ESSR, and peak load for each placement, and then evaluates the performance with the use of a fitness function (refer to Eqs. (22)~(25)).

3.1. Multi-objective optimization results for the cluster considering SBESS placement

Table 5 and Fig. 4 show the performance evaluation results of optimization objectives according to the optimal placement of SBESS in each cluster.

In order to analyze the effects of SBESS placement, the analysis was conducted based on the three cases established as follows: (i) Case A: case without SBESS installation; (ii) Case B: case with optimal SBESS placement; (iii) Case C: case with SBESS installed in all buildings.

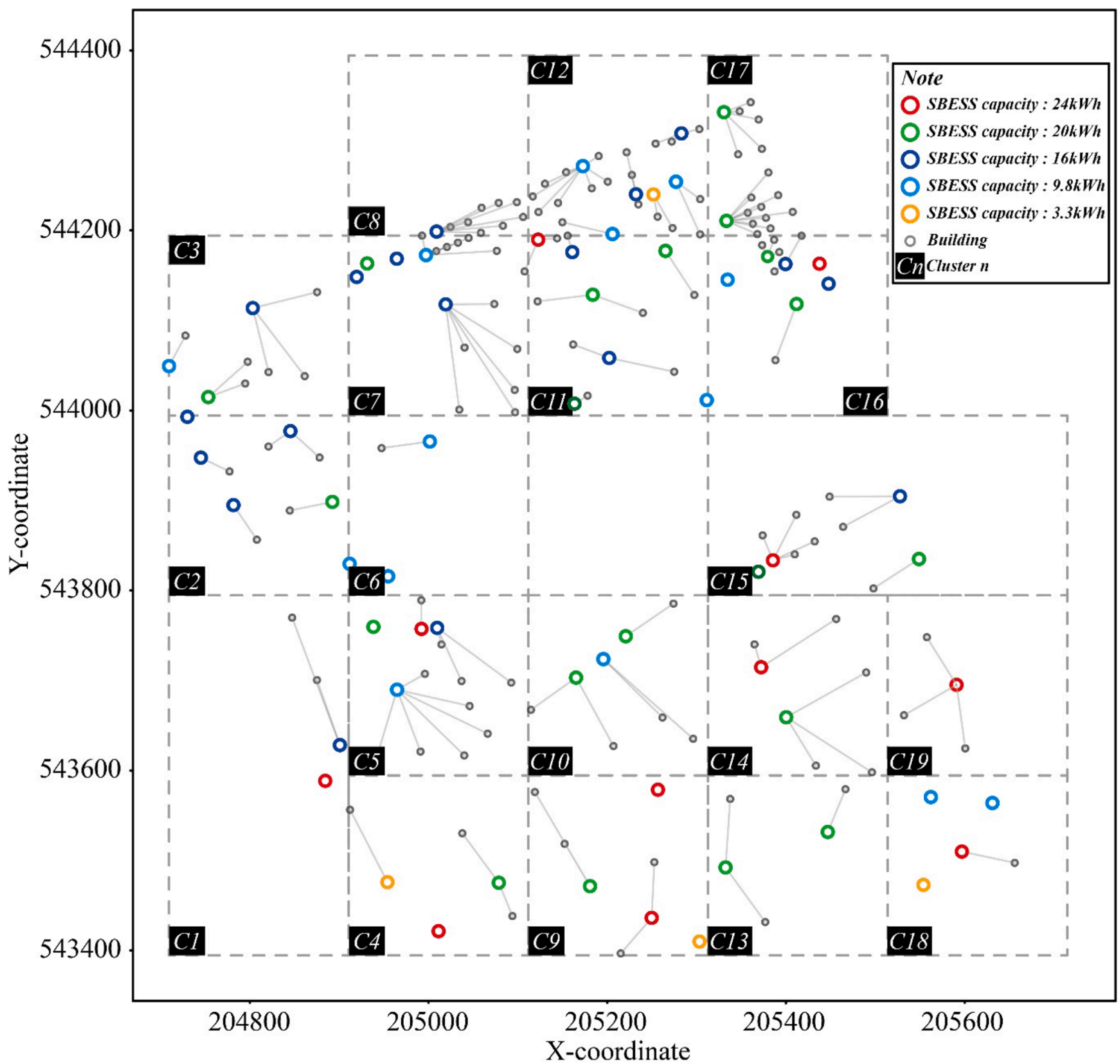
Table 6 summarizes the results of economic performance (i.e., NPV) analysis for each cluster according to the SBESS placement strategies. The NPV analysis found that in the absence of SBESS (case A), most clusters showed positive NPV (i.e., USD 32,694.63 ~ USD 197,323.46). However, in the presence of SBESS, NPV decreased in all clusters, showing negative values. This suggests that the current SBESS installation, operation and maintenance costs significantly exceed the electricity bill saving effects, or that the benefits obtained from the self-consumption of electricity produced by PV is very low compared to the gain from the sale to the external electricity grid.

When comparing case B and case C, the results were generally favorable for case B. In many clusters, such as cluster 12 and cluster 19, case B demonstrated better economic performance than case C (e.g., USD -638,256.10 to USD -709,533.26 in cluster 12, and USD -284,713.65 to USD -335,593.56 in cluster 19). However, there were some clusters, such as cluster 11, where case C showed marginally better NPV values than case B (e.g., USD -538,878.16 to USD -521,943.69 in cluster 11). These variations suggest that the economic outcomes of the proposed strategy can be influenced by cluster-specific factors, such as energy consumption patterns, building density, and local installation costs.

While the economic performance analysis reveals that case B does not consistently outperform case C in all clusters, it generally achieves comparable or slightly better results in many cases. For example, in cluster 1 and cluster 5, case B demonstrated slightly higher NPV compared to case C, effectively minimizing unnecessary installations and associated costs. However, in some clusters, such as cluster 11, case C showed marginally better economic performance. These variations are expected, as the optimization process balances multiple objectives, and economic performance alone may not always be maximized.

Table 7 summarizes the results of technical and environmental performance (i.e., ESSR, and peak load) for each cluster according to the SBESS placement strategies. First, in terms of ESSR, the installation of SBESS showed a remarkable improvement in all clusters (refer to Table 7). When compared to the case with no SBESS installed (case A), where ESSR was 0%, the ESSR increased to 2.97% up to 17.44% depending on the cluster when SBESS was installed. This confirms a further improvement in the self-consumption of electricity produced by PV. Moreover, in most clusters, the difference in ESSR between the optimal SBESS placement (case B) and the installation of SBESS in all buildings (case C) was insignificant (i.e., ±0.03%). However, in some clusters, the optimal SBESS placement showed higher ESSR than the installation of SBESS in all buildings. This indicates that the optimal placement of SBESS can be as effective or even more effective than the installation of SBESS in all buildings in increasing energy independence in a city. Even in terms of peak load, the installation of SBESS had a positive effect. In all clusters, the peak load decreased after the installation of SBESS (refer to Table 7). Particularly in some clusters, there was a significant improvement with the peak load decreased by up to 37.19% (i.e., 451.2kWh to 283.4kWh for cluster 17). Overall, the peak load difference between the optimal SBESS placement (case B) and the installation of SBESS in all buildings (case C) was less than ±1% in most clusters. This suggests that optimal placement strategies can provide similar performance in peak load reduction compared to cases with SBESS installed in all buildings, while being more effective at reducing the peak load in some clusters.

The research results clearly demonstrate the complexity and multifacetedness of SBESS introduction. Particularly notable is the effectiveness of optimal placement strategies for SBESS. In most cases, the



Note: The coordinates system are expressed in meters based on the Korea Central Belt.
The X-axis represents the west-east direction, while the Y-axis represents the south-north direction.
On the coordinate plane, 1 unit represents 1 meter.

Fig. 4. Optimal placement results of SBESS for all cluster.

optimal placement of SBESS can achieve similar or better performance than the installation of SBESS in all building even with a smaller number of SBESS. This highlights the importance of establishing cost-effective SBESS placement strategies and suggests a method to achieve maximum efficiency with a limited budget.

In terms of economic performance, the installation of SBESS results in a decrease in NPV, but its advantages, such as an increase in ESSR and a reduction in peak load, are clear in terms of technical and environmental performance. In particular, it was confirmed that the optimal placement of SBESS can provide similar or better technical and environmental performance, while minimizing economic losses, compared to the installation of SBESS in all buildings. In cluster 17, the optimal placement of SBESS helped to achieve a slightly higher ESSR (i.e., case B: 4.51% and case C: 4.48%) than the installation of SBESS in all buildings. Also, in cluster 11, the optimal placement of SBESS showed a lower peak load (i.e., case B: 347.46kWh and case C: 350.87kWh) than the installation of SBESS in all buildings.

The difference in performance between clusters suggests that site-specific characteristics (e.g., electricity consumption pattern, PV electricity generation, electricity billing system, etc.) can have a significant impact on the performance of SBESS. For example, the high ESSR improvement of cluster 19 may be related to the cluster's unique energy demand-supply pattern, which suggests that a detailed consideration of various site-specific characteristics is needed when SBESS placement strategies are set up. Therefore, it is important to develop optimized SBESS placement and operation strategies tailored to the characteristics of each region and building.

In addition, when SBESS-related policies are set up, measures not only to improve technical and environmental performance but also to enhance economic performance should be taken into account. While the current analysis shows negative NPV values in all cases, this does not mean that SBESS installations are inherently unsuitable. Rather, the results reflect the high upfront costs of SBESS and the limited financial returns under current market conditions. In this context, it may be necessary to promote technical innovation to reduce SBESS installation costs, electricity market structure improvement, or government subsidy payments. Furthermore, it may be necessary to develop SBESS capacity calculations and operation strategies considering site-specific characteristics, such as electricity consumption patterns, renewable energy production, and electricity billing systems. Over time, advancements in

Table 6
Analysis of economic performance by cluster according to SBESS placement strategies.

Cluster ID	Net present value (USD)		
	Case A	Case B	Case C
1	82,741.90	-322,134.11	-322,590.47
2	114,127.89	-501,199.90	-501,656.21
3	136,344.57	-487,120.95	-498,466.32
4	142,520.33	-531,644.24	-544,080.99
5	169,463.36	-632,275.84	-627,539.89
6	56,231.60	-304,278.44	-300,593.42
7	188,370.00	-821,469.37	-826,340.51
8	32,694.63	-163,249.56	-178,788.46
9	117,752.22	-484,829.16	-476,271.73
10	197,323.46	-741,780.22	-759,774.35
11	87,863.26	-538,878.16	-521,943.69
12	141,580.68	-638,256.10	-709,533.26
13	97,216.65	-358,821.84	-358,479.53
14	189,254.42	-719,358.79	-717,709.86
15	189,303.50	-725,029.68	-734,494.76
16	111,630.05	-584,262.89	-581,842.55
17	54,975.59	-353,131.25	-386,016.18
18	89,596.35	-330,977.89	-330,179.19
19	74,872.20	-284,713.65	-335,593.56

Note: The exchange rate (KRW/USD) is 1383.11 won to a U.S. dollar (as of 23 July 2024), Case A stand for case without SBESS installation, Case B stand for case with optimal SBESS placement and Case C stand for case with SBESS installed in all buildings.

energy storage technologies and supportive policy measures could make SBESS installations more economically viable, aligning them with broader goals of energy sustainability and resilience.

In conclusion, the results of this study show that SBESS has the potential to contribute to improving the efficiency and stability of renewable energy systems. But at the same time, it suggests that there are challenges to be overcome under the current economic conditions. However, in the future, the economic feasibility of SBESS is expected to improve with technological advancements, installation cost reductions, and electricity market changes. In this manner, a sensitivity analysis on electricity prices and SMP and REC fluctuations will be conducted in the following chapter. The analysis evaluates how sensitively the economic feasibility of SBESS responds to electricity price and SMP and REC fluctuations, and explores under what conditions SBESS can be economically feasible. This will help evaluate the long-term feasibility of SBESS introduction and contribute to further refining the optimal placement strategies considering site-specific characteristics.

3.2. Sensitivity analysis of energy cost on technical, environmental, and economic performance of SBESS

In Section 3.1, it was confirmed that the optimal SBESS placement strategies are effective in terms of technical and environmental performance. However, it was also found that under the current electricity market prices (i.e., electricity price, SMP, and REC price), the installation of SBESS results in a large decrease in NPV, making it difficult to ensure economic feasibility. However, the electricity market is constantly changing. The electricity market price volatility can have a particularly significant impact on the economic performance of SBESS. Therefore, in this section, a sensitivity analysis is to be conducted according to SMP and REC prices, and electricity price fluctuations. The objective of the sensitive analysis is to evaluate the economic feasibility of SBESS under various price scenarios and to determine under what conditions the installation of SBESS can be economically advantageous. Furthermore, it aims to quantify how sensitive the economic performance of SBESS is to electricity market price fluctuations and to predict the potential value of SBESS according to future energy policy and market changes.

A sensitivity analysis conducted on all clusters is extremely costly in terms of time and resources. However, the expected effects are insignificant compared to the time required for the analysis. Therefore, in this study, a sensitivity analysis is to be conducted on cluster 12, which has the largest number of buildings belonging to the cluster. Fig. 5 shows the sensitivity analysis results when the electricity price increases threefold, and the SMP and REC prices are doubled. The multiplier of the electricity market price was set up by referring to the difference of the existing electricity market price.

Prior to performing the sensitive analysis, SBESS was set to discharge during peak hours and charge as much as possible during non-peak hours, given the initial situation in which the electricity price was low compared to the SMP and REC prices. Against this backdrop, the analysis results confirmed that in the current PV-SBESS system, electricity produced by PV is mainly used for self-consumption.

The sensitive analysis results showed that electricity price fluctuations had the greatest impact on the performance of the PV-SBESS system (refer to Fig. 5). NPV significantly increased from USD -622,209.86 to USD 522,883.60 with the increasing electricity prices. This shows that the economic value of SBESS increases significantly when electricity prices increase. In particular, the conversion of NPV from a negative value to positive means that through the SBESS installation, it is possible to ensure economic feasibility when the electricity price rises above a certain level. Meanwhile, ESSR increased from 2.73% to 3.95% as the electricity price increased three times. This indicates that SBESS can contribute to improving energy independence in energy communities. However, this increase rate is relatively low, suggesting that there is still a high level of dependence on the external electricity grid. On the other hand, the peak load showed a tendency to increase slightly from 660.93 kWh to 662.09 kWh as the electricity price increased. This is because when the electricity price increased, the number of SBESS installed in the cluster decreased from 18 to 4, resulting in a reduction in discharge capacity during peak hours.

Meanwhile, SMP and REC price fluctuations had a relatively insignificant effect on all indicators. This suggests that the impact of selling electricity to the external electricity grid on overall economic efficiency is limited in the relationship among the current PV-SBESS system configuration, SBESS operation logic, PV electricity generation, and electricity consumption. However, given the very low PV electricity generation (i.e., 303,917.74kWh) compared to the annual electricity consumption (i.e., 5420,681kWh), this finding can be seen as a meaningful result.

Based on the sensitivity analysis results of this study, the following implications and future research directions can be derived:

- Importance of electricity market design (An et al., 2021, 2022): Given the significant impact of electricity prices on the performance of the PV-SBESS system and its economic performance, policy makers should carefully devise electricity pricing policies to promote the expansion of SBESS. In this regard, it is worthwhile considering identifying the point in time at which NPV turns from a negative value into a positive one and linking it with appropriate subsidy payment policies. This will contribute to improving the resilience and sustainability of the PV-SBESS system in the long term.
- Necessity of expending renewable energy deployment (An et al., 2019, Kim et al., 2023): The limited increase of ESSR suggests that the current PV electricity generation is significantly low compared to the electricity consumption of the energy community. In this study, PV systems were installed on building rooftops, given the realistic constraints. However, the installation of additional renewable energy systems is needed to achieve urban energy independence. In this connection, urban planning and legal regulations should be taken into account.
- Optimization of SBESS operation strategy (An and Hong, 2025): Considering the trade-off relationship between economic performance, and technical and environmental performance, a multifaceted approach needs to be taken to determine the number of SBESS

Table 7
Analysis of technical and environmental performance by cluster according to SBESS placement strategies.

Cluster ID	Electricity self-sufficiency rate (%)			Peak load (kWh)		
	Case A	Case B	Case C	Case A	Case B	Case C
1	0	4.95	4.95	308.0	291.22	291.15
2	0	5.00	5.00	454.3	425.06	424.99
3	0	3.33	3.32	789.2	749.35	750.07
4	0	7.58	7.57	406.3	333.17	331.29
5	0	4.36	4.36	716.4	583.42	586.05
6	0	4.27	4.27	356.5	305.91	308.37
7	0	8.20	8.19	586.2	439.96	443.62
8	0	3.28	3.26	287.8	215.99	215.99
9	0	10.52	10.53	249.6	211.16	212.42
10	0	5.31	5.3	855.8	668.81	666.95
11	0	5.74	5.75	420.4	347.46	350.87
12	0	3.94	3.91	939.7	674.5	670.99
13	0	8.74	8.74	240.5	212.82	212.87
14	0	4.72	4.72	888.7	659.83	663.35
15	0	2.97	2.97	1502.5	1102.09	1102.94
16	0	4.27	4.27	598.6	481.05	482.11
17	0	4.51	4.48	451.2	283.4	283.4
18	0	7.96	7.96	221.5	196.15	196.27
19	0	17.44	17.26	122.8	92.39	80.13

Note: Case A stand for case without SBESS installation, Case B stand for case with optimal SBESS placement and Case C stand for case with SBESS installed in all buildings

installations and their capacities. In particular, it is necessary to derive the optimal operation strategy via simulation and optimization for various SEBSS operation strategies.

- PV-SBESS system design from a long-term perspective: The current impact of SMP and REC prices is insignificant. However, in the future, the impact may increase as renewable energy supply expands. Therefore, there is a need to devise a flexible PV-SBESS system design considering the interaction with the electricity market from a long-term perspective. This will expand revenue generation opportunities for energy communities and enable efficient integration with the entire electric system.

In conclusion, the PV-SBESS system has great potential in terms of improving the sustainability and energy independence of energy communities. However, in order to realize its full potential, efforts need to be made in various aspects, such as technology development, policy-making, and social acceptance. In future studies, the development of a comprehensive framework is needed to integrate these multifaceted approaches, which will ultimately contribute to building a more efficient and sustainable PV-SBESS system.

4. Conclusion

This study aims to derive the optimal placement (i.e., location and capacity) of SBESS in consideration of various optimization objectives (i.e., NPV, ESSR, and peak load) when installing and sharing BESS in the energy community. To this end, a five-step process was performed: (i) selection of the study area; (ii) data collection; (iii) PV-SBESS system modeling; (iv) definition of the optimization goal for the optimal placement of SBESS; and (v) optimal SBESS placement using multi-objective optimization. The major findings of this study are as follows.

- The analysis revealed that NPV was negative across all clusters, ranging from USD -821,469.37 to USD -163,249.56. This indicates that current SBESS installation and operation costs exceed the economic benefits gained from electricity self-consumption or sales under existing conditions.
- Regarding technical and environmental performance, ESSR ranged from 2.97% to 17.44%, and peak load reductions varied between 92.39kWh and 1,102.09kWh depending on the cluster. The performance differences between optimal SBESS placement and full placement were not significant (ESSR: $\pm$0.03%, peak load: $\pm$1%), emphasizing the importance of cost-effective placement strategies.
- The analysis results for each cluster were not linear but different from one another. This indicates that site-specific characteristics can have a significant impact on the performance of SBESS.
- The sensitivity analysis results indicate that electricity prices have a significant impact on the economic feasibility of SBESS, while the effects of SMP and REC are limited. These findings highlight the importance of electricity market design and the expansion of renewable energy installations to enhance the adoption and sustainability of SBESS.

The methodology proposed in this study is superior in placing SBESS in energy communities in the following aspects: (i) A multi-objective optimization was adopted to comprehensively consider trade-offs among economic, technical, and environmental performance. This helps to derive the optimal placement of SBESS, which is balanced in various aspects without being biased toward a single objective. This methodology can provide a more suitable solution for designing complex energy systems in real life; (ii) The data of the actual energy community were collected and reflected to conduct the analysis, increasing the practicality and reliability of the research results. This study was based on the actual data, not a virtual electricity network, and the results of this study can provide useful insights for policy-making for SBESS expansion, technology development and academic research directions; (iii) In addition, a variety of practical factors were considered to enhance the reality of the model. Transmission losses due to the distance between the SBESS, PV, and building, capacity deterioration of PV and SBESS with time, and self-discharge, and charge/discharge efficiency of SBESS were included in considerations for the model. These detailed considerations are expected to greatly enhance the practicality and applicability of the research results.

Despite its contributions, this study possesses the following limitations. First, this study comprehensively considered three optimization objectives, assigning identical weight factors to all. This approach provides a balanced and intuitive basis for multi-objective optimization; however, it does not fully account for the diverse characteristics of energy communities that could influence optimization outcomes. For example, demographic factors such as the age composition of community members, which can significantly affect energy consumption patterns and peak load timings, were not explicitly considered. Incorporating such characteristics into the optimization process could allow for the exploration of dynamic weight factor settings that better reflect the unique attributes of energy communities and enhance the applicability of the proposed methodology. Second, data for the past several years were collected and used to reflect various variables such as estimation of PV electricity generation and changes in economic indexes (i.e., real discount rate, electricity market price, etc…) as realistically as possible. However, the use of fixed values has clear limitations in predicting and analyzing the future. In this regard, more diverse future scenarios should be considered through the estimation of PV electricity generation via climate change predictions, and the integration of models for predicting the rate of increase in electricity market prices. Third, in evaluating the economic performance, this study considered various economic factors for SBESS, such as initial investment costs, maintenance costs, transmission losses, and revenue from electricity sales. However, incorporating additional revenue factors, such as voltage deviation improvement cost, could further enhance the economic feasibility of SBESS. Finally, a rule-based SBESS operation logic based on the electricity market price was used to derive the optimal placement of the PV-SBESS system. This can provide a practical and intuitive approach but may fail to fully optimize the potential performance of the PV-SBESS system. Therefore, further research is needed to develop an algorithm capable of dynamically optimizing the scheduling of SBESS and integrate it with the process of placement optimization.

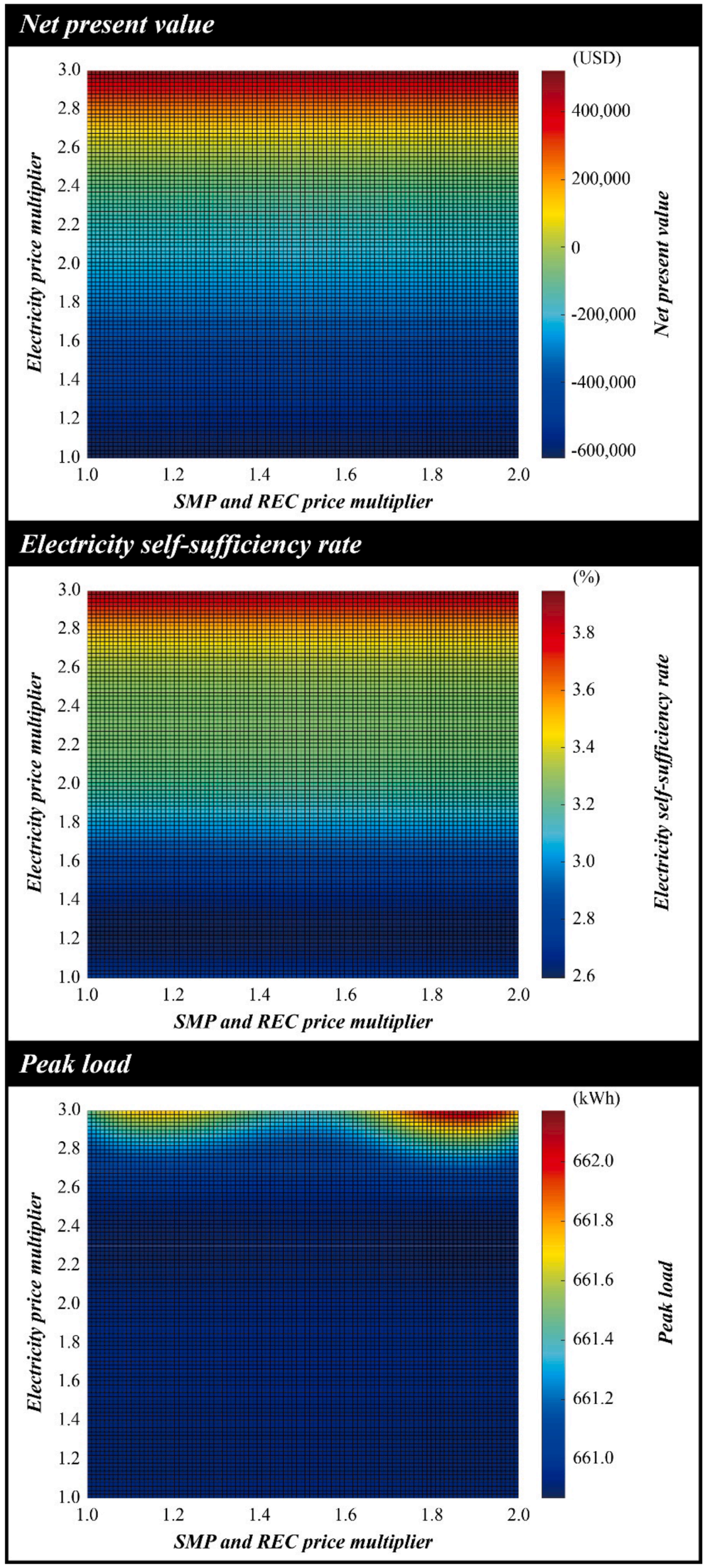


Fig. 5. Results of sensitivity analysis on the changes in electricity market price.

CRediT authorship contribution statement

Jongbaek An: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Taehoon Hong:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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